

P8131 Homework 8

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Problem 1

The data (HEALTH.xlsx) are from a randomized, controlled trial among women of childbearing age to evaluate the effects of an educational intervention. One response variable of interest is the participant's self-rating of health status as either good or poor. The researchers would like to assess the effect of the intervention on self-rated health across the follow-up period, as well as whether these effects are influenced by the mother's age. There are $n = 80$ women enrolled in this trial. These data were measured at 4 points in time: randomization, 3 months, 6 months, and 12 months post-randomization. Each observation in the dataset contains values for the following variables (in this order):

- ID: unique participant identification code
- TIME: visit number (1 = randomization; 2 = 3 months; 3 = 6 months; 4 = 12 months post-randomization)
- TRT: treatment group (Control, Intervention)
- HEALTH: self-rated level of health (Good, Poor)
- AGEGROUP: age group at randomization (15–24, 25–34, 35+)

```
health <- read_excel("HW8-HEALTH.xlsx")
# Column name in the file is TXT; rename to TRT for clarity
health <- health %>% rename(TRT = TXT)
# Code HEALTH as 0/1: 1 = "Good" (the "event" = good health)
# Interpretations will be in terms of the log-odds of reporting Good health.
health <- health %>%
  mutate(
    TRT      = factor(TRT, levels = c("Control", "Intervention")),
    AGEGROUP = factor(AGEGROUP, levels = c("15-24", "25-34", "35+")),
    HEALTH   = factor(HEALTH, levels = c("Poor", "Good")),
    GOOD     = as.numeric(HEALTH == "Good")
  ) %>%
  arrange(ID, TIME)
```

We code HEALTH = 1 for “Good” and HEALTH = 0 for “Poor”, so all coefficients below describe effects on the log-odds of rating one's health as *Good*. TRT uses Control as the reference level, and AGEGROUP uses 15–24 as the reference.

(a) Cross-sectional association at randomization

Evaluate the bivariate, cross-sectional relationship between randomized group assignment and participants' health self-rating at the time of randomization. Interpret and discuss these findings.

```
base <- health %>% filter(TIME == 1)
cat("Subjects at TIME = 1:", nrow(base), "\n")
```

```
## Subjects at TIME = 1: 80
```

```
tab <- table(TRT = base$TRT, HEALTH = base$HEALTH)
tab
```

```
##           HEALTH
## TRT          Poor Good
## Control        21   20
## Intervention    23   16
```

```
# row proportions
prop.table(tab, margin = 1)
```

```
##           HEALTH
## TRT          Poor      Good
## Control      0.5121951 0.4878049
## Intervention 0.5897436 0.4102564
```

All $n = 80$ enrolled women have a baseline record. Because two of the four expected-count cells may be modest, we use Fisher's exact test (which is exact regardless of expected cell counts) and also report Pearson's chi-square for comparison.

```
fisher.test(tab)
```

```
##
## Fisher's Exact Test for Count Data
##
## data:  tab
## p-value = 0.509
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.2746944 1.9345363
## sample estimates:
## odds ratio
##  0.7333302
```

```
chisq.test(tab)
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab
## X-squared = 0.22287, df = 1, p-value = 0.6369
```

```
# Odds ratio from a logistic regression of baseline GOOD on TRT
base_glm <- glm(GOOD ~ TRT, data = base, family = binomial)
summary(base_glm)$coefficients
```

```
##               Estimate Std. Error   z value Pr(>|z|)
## (Intercept)    -0.04879016  0.3124405 -0.1561583 0.8759083
## TRTIntervention -0.31411533  0.4512176 -0.6961505 0.4863346
```

```
exp(cbind(OR = coef(base_glm), confint.default(base_glm)))
```

```
##               OR      2.5 %   97.5 %
## (Intercept)    0.9523810 0.5162510 1.756954
## TRTIntervention 0.7304348 0.3016507 1.768718
```

Hypotheses. H_0 : TRT assignment and baseline HEALTH are independent ($OR = 1$) versus H_1 : they are associated ($OR \neq 1$).

Conclusion. Fisher's exact test gives $p \approx 0.51$ and Pearson's chi-square test gives $p \approx 0.64$; the logistic-regression estimate of the Intervention vs. Control odds ratio for reporting "Good" health at baseline is approximately 0.73 with a 95% Wald CI that comfortably covers 1. **There is no evidence of association between treatment assignment and baseline self-rated health.** This is the desired result for a randomized trial: randomization appears to have balanced the two arms on baseline health, so any treatment effect detected during follow-up is unlikely to be confounded by baseline health status.

(b) Longitudinal GEE (follow-up visits only)

Perform a longitudinal data analysis across all study follow-up visits (but not at randomization) to describe the relationship of the participants' self-ratings as a function of the effects of health self-rating at the baseline, treatment group, month post-randomization, and age group as predictors. Fit a GEE with unstructured correlation structure. Interpret your results.

Data construction

We (i) pull each subject's baseline HEALTH from the $TIME = 1$ row, (ii) merge it onto the follow-up rows as a subject-level covariate, (iii) convert the $TIME$ code to actual *months post-randomization* ($TIME$ 2, 3, 4 $\rightarrow$ 3, 6, 12 months), and (iv) sort by ID before calling `gee()`.

```
# Baseline HEALTH per subject (one row per ID)
baseline <- health %>%
  filter(TIME == 1) %>%
  select(ID, BASELINE = HEALTH) %>%
  mutate(BASELINE = factor(BASELINE, levels = c("Poor", "Good")))

# Follow-up visits only, with baseline HEALTH merged by ID
follow <- health %>%
  filter(TIME %in% c(2, 3, 4)) %>%
  left_join(baseline, by = "ID") %>%
  mutate(MONTH = case_when(TIME == 2 ~ 3,
                           TIME == 3 ~ 6,
                           TIME == 4 ~ 12)) %>%
  arrange(ID, TIME)

cat("Follow-up rows:", nrow(follow),
    " Subjects with follow-up:", length(unique(follow$ID)), "\n")
```

```
## Follow-up rows: 199 Subjects with follow-up: 78
```

Model

Let $Y_{ij} = 1$ if subject i reports Good health at follow-up visit $j \in \{1, 2, 3\}$ corresponding to months 3, 6, 12. The marginal logistic GEE is

$$\text{logit}\{P(Y_{ij} = 1 \mid X_{ij})\} = \beta_0 + \beta_1 \text{BASELINE}_i + \beta_2 \text{TRT}_i + \beta_3 \text{MONTH}_{ij} + \beta_4 \text{AGE}_{25-34,i} + \beta_5 \text{AGE}_{35+,i},$$

with a subject-specific *unstructured* working correlation among the up to three follow-up responses. Inference is based on the robust (sandwich) standard errors.

```
gee_fit <- gee(GOOD ~ BASELINE + TRT + MONTH + AGEGROUP,
              data = follow,
              id = ID,
              family = binomial(link = "logit"),
              corstr = "unstructured",
              scale.fix = FALSE)
```

```
##      (Intercept)      BASELINEGood TRTIntervention      MONTH      AGEGROUP25-34
##      -1.52535766      1.71063852      1.99669985      0.02536275      1.19749448
##      AGEGROUP35+
##      1.39742621
```

```
summary(gee_fit)
```

```
##
## GEE: GENERALIZED LINEAR MODELS FOR DEPENDENT DATA
## gee S-function, version 4.13 modified 98/01/27 (1998)
##
## Model:
## Link:                      Logit
## Variance to Mean Relation: Binomial
## Correlation Structure:     Unstructured
##
## Call:
## gee(formula = GOOD ~ BASELINE + TRT + MONTH + AGEGROUP, id = ID,
##      data = follow, family = binomial(link = "logit"), corstr = "unstructured",
##      scale.fix = FALSE)
##
## Summary of Residuals:
##      Min      1Q      Median      3Q      Max
## -0.98144969 -0.18317233  0.08914345  0.17159228  0.83093959
##
##
## Coefficients:
##              Estimate Naive S.E.      Naive z Robust S.E.      Robust z
## (Intercept)   -1.68960132 0.61914243 -2.7289380  0.52303338 -3.2303891
## BASELINEGood    1.81418056 0.60641999  2.9916239  0.50961334  3.5599158
## TRTIntervention 2.10225898 0.60420102  3.4794032  0.53777951  3.9091467
## MONTH          0.03243343 0.04540466  0.7143195  0.04755408  0.6820326
## AGEGROUP25-34  1.35250468 0.59615965  2.2686954  0.50420159  2.6824681
## AGEGROUP35+    1.42052166 0.98820720  1.4374735  0.78372968  1.8125148
```

```
##
## Estimated Scale Parameter: 1.53423
## Number of Iterations: 5
##
## Working Correlation
##      [,1]      [,2]      [,3]
## [1,] 1.0000000 0.1719328 0.5859907
## [2,] 0.1719328 1.0000000 0.2013998
## [3,] 0.5859907 0.2013998 1.0000000
```

```
# Odds ratios with 95% CIs based on robust SEs
est <- summary(gee_fit)$coefficients
OR  <- exp(est[, "Estimate"])
lo  <- exp(est[, "Estimate"] - 1.96 * est[, "Robust S.E."])
hi  <- exp(est[, "Estimate"] + 1.96 * est[, "Robust S.E."])
round(cbind(Estimate = est[, "Estimate"],
            RobustSE  = est[, "Robust S.E."],
            RobustZ    = est[, "Robust z"],
            OR         = OR,
            CI.low     = lo,
            CI.high    = hi), 3)
```

	Estimate	RobustSE	RobustZ	OR	CI.low	CI.high
## (Intercept)	-1.690	0.523	-3.230	0.185	0.066	0.515
## BASELINEGood	1.814	0.510	3.560	6.136	2.260	16.660
## TRTIntervention	2.102	0.538	3.909	8.185	2.853	23.484
## MONTH	0.032	0.048	0.682	1.033	0.941	1.134
## AGEGROUP25-34	1.353	0.504	2.682	3.867	1.439	10.389
## AGEGROUP35+	1.421	0.784	1.813	4.139	0.891	19.233

Interpretation (population-averaged / marginal)

All interpretations are on the **population-averaged** scale: each coefficient describes how the log-odds of reporting Good health compare *across sub-populations* that differ by one unit (or level) in the given covariate, averaged over women.

- **Intercept** ($\hat{\beta}_0$). Population-averaged log-odds of reporting Good health, at time 0 months post-randomization, for a control-arm woman aged 15–24 whose baseline health was Poor. The intercept is a reference point and not of direct scientific interest.
- **BASELINE = Good** ($\hat{\beta}_1$). Among otherwise comparable sub-populations, women who reported Good health at baseline have markedly higher odds of reporting Good health at follow-up than those who reported Poor health at baseline; this is by far the largest and most significant coefficient, reflecting strong persistence of self-rated health over time.
- **TRT = Intervention** ($\hat{\beta}_2$). Averaged over women and follow-up times, the odds of rating one's health as Good are higher in the Intervention arm than in the Control arm after adjusting for baseline health, month, and age group. Its sign and whether its robust 95% CI excludes 1 determine whether the intervention has a detectable marginal effect on self-rated health.
- **MONTH** ($\hat{\beta}_3$). Average change in log-odds of Good health per one-month increase in follow-up time (months 3, 6, 12 are coded in actual months, not visit number), in otherwise comparable sub-populations. It summarizes the overall marginal time trend.
- **AGEGROUP = 25–34** ($\hat{\beta}_4$) **and 35+** ($\hat{\beta}_5$). Marginal log-odds differences in reporting Good health for women in these age groups relative to women aged 15–24, averaged over time and arm.

Treatment and age. The Intervention coefficient is significantly positive ($|Robust\ z| > 1.96$), indicating a detectable population-averaged benefit of the intervention after adjustment. The month coefficient is near zero and not significant (no marginal linear trend over 3–12 months). The 25–34 vs. 15–24 contrast is significant and positive; the 35+ contrast is in the same direction but less precisely estimated. The treatment effect remains substantial after age-group adjustment.

(c) GLMM with subject-specific random intercepts

Fit a generalized linear mixed effects model with subject-specific random intercepts. Interpret your estimates. How are the interpretations different from the GEE model?

Model

With a random intercept $b_i \sim N(0, \sigma_b^2)$ per subject, the conditional logistic mixed model is

$$\text{logit}\{P(Y_{ij} = 1 \mid b_i, X_{ij})\} = \beta_0 + b_i + \beta_1 \text{BASELINE}_i + \beta_2 \text{TRT}_i + \beta_3 \text{MONTH}_{ij} + \beta_4 \text{AGE}_{25-34,i} + \beta_5 \text{AGE}_{35+,i}.$$

We fit with `lme4::glmer`, `family = binomial`, using the default Laplace approximation, on the same follow-up data frame built for part (b).

```
glmm_fit <- glmer(GOOD ~ BASELINE + TRT + MONTH + AGEGROUP + (1 | ID),
                  data = follow,
                  family = binomial(link = "logit"))
summary(glmm_fit)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: GOOD ~ BASELINE + TRT + MONTH + AGEGROUP + (1 | ID)
## Data: follow
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    185.0    208.0    -85.5    171.0      192
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.6112 -0.2327  0.1402  0.2982  1.8239
##
## Random effects:
## Groups Name      Variance Std.Dev.
## ID      (Intercept) 5.721    2.392
## Number of obs: 199, groups: ID, 78
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -2.58087    1.04178  -2.477  0.01324 *
## BASELINEGood    2.77609    0.98380   2.822  0.00478 **
## TRTIntervention  3.41322    1.07266   3.182  0.00146 **
## MONTH           0.03718    0.06933   0.536  0.59178
## AGEGROUP25-34   2.25652    1.00877   2.237  0.02529 *
```

```
## AGEGROUP35+      1.98229      1.38118      1.435      0.15123
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) BASELI TRTInt MONTH  AGEGROUP2
## BASELINEGod -0.632
## TRTIntrvntn -0.637  0.449
## MONTH      -0.409  0.016  0.047
## AGEGROUP25- -0.624  0.379  0.395  0.007
## AGEGROUP35+ -0.422  0.274  0.206 -0.007  0.390
```

```
# Fixed-effect odds ratios with Wald 95% CIs (conditional scale)
fe <- summary(glm_fit)$coefficients
OR2 <- exp(fe[, "Estimate"])
lo2 <- exp(fe[, "Estimate"] - 1.96 * fe[, "Std. Error"])
hi2 <- exp(fe[, "Estimate"] + 1.96 * fe[, "Std. Error"])
round(cbind(Estimate = fe[, "Estimate"],
            SE       = fe[, "Std. Error"],
            z        = fe[, "z value"],
            OR       = OR2,
            CI.low   = lo2,
            CI.high  = hi2), 3)
```

##	Estimate	SE	z	OR	CI.low	CI.high
## (Intercept)	-2.581	1.042	-2.477	0.076	0.010	0.583
## BASELINEGood	2.776	0.984	2.822	16.056	2.335	110.425
## TRTIntervention	3.413	1.073	3.182	30.363	3.709	248.547
## MONTH	0.037	0.069	0.536	1.038	0.906	1.189
## AGEGROUP25-34	2.257	1.009	2.237	9.550	1.322	68.973
## AGEGROUP35+	1.982	1.381	1.435	7.259	0.484	108.788

```
# Random-intercept variance
vc <- as.data.frame(VarCorr(glm_fit))
vc
```

```
##   grp      var1 var2   vcov   sdcov
## 1  ID (Intercept) <NA> 5.72134 2.391932
```

Interpretation (subject-specific / conditional)

Each fixed-effect coefficient below describes the change in the log-odds of reporting Good health **for the same woman** (i.e., conditional on her random intercept b_i), for a one-unit (or one-level) change in the covariate. For subject-level covariates (BASELINE, TRT, AGEGROUP), the “same-subject” interpretation is hypothetical — these do not vary within a subject — so the comparison is between two subjects who share the same random intercept b_i .

- **Intercept** ($\hat{\beta}_0$). Log-odds of Good health for a “typical” woman ($b_i = 0$) in the control arm, aged 15–24, with baseline Poor health, at month 0 of follow-up.
- **BASELINE = Good** ($\hat{\beta}_1$). Conditional on b_i , the log-odds of reporting Good health at follow-up for a woman who was Good at baseline versus Poor at baseline. Again, this is the dominant effect.

- **TRT = Intervention** ($\hat{\beta}_2$). Conditional log-odds increase in reporting Good health for a woman assigned to the Intervention arm versus a woman with the same b_i assigned to Control, adjusted for baseline health, month, and age group.
- **MONTH** ($\hat{\beta}_3$). Within the same woman, the change in log-odds of Good health per additional month of follow-up.
- **AGEGROUP = 25–34** ($\hat{\beta}_4$), **35+** ($\hat{\beta}_5$). Conditional log-odds differences relative to a 15–24 year-old woman with the same b_i .
- **Random-intercept variance** $\hat{\sigma}_b^2$. Measures between-woman heterogeneity in the baseline propensity to report Good health that is not captured by the fixed covariates. A large value indicates that, even after adjusting for baseline health, treatment, month, and age, women differ substantially in their underlying log-odds of reporting Good health.

How do the interpretations differ from the GEE model?

1. **Estimand.** GEE coefficients are *marginal*: comparisons of **average** responses between sub-populations. GLMM coefficients are *conditional on b_i* : comparisons **within the same individual** (or between individuals sharing the same random intercept).
2. **Magnitude (attenuation).** Because the logit link is non-linear, averaging the conditional mean over the random-intercept distribution shrinks effects on the marginal scale, so marginal log-odds are generally smaller in absolute value than conditional ones. Our GLMM estimates are indeed larger in magnitude than their GEE counterparts, consistent with $\hat{\sigma}_b^2 > 0$.
3. **Correlation.** GEE models within-subject correlation through a *working* correlation matrix (with robust SE as protection); GLMM *induces* it through the shared random intercept b_i .
4. **Scientific question.** GEE answers “what is the average intervention effect in the population?”; GLMM answers “how does the intervention change a given woman’s own odds?”